

CLINTON, IA, USA

WMO: 725473

Lat: 41.833N

Lon: 90.333W

Elev: 700

StdP: 14.33

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 94979

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -8.2	(c) -1.8	(d) -17.2	(e) 2.2	(f) 0.4	(g) -10.6	(h) 3.2	(i) 2.4	(j) 31.6	(k) 24.2	(l) 28.2	(m) 18.3	(n) 7.9	(o) 280	(p) 0.580

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.1	90.6	75.2	88.3	74.2	85.4	72.8	78.4	87.4	76.7	85.1	74.8	82.4	12.0	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.4	136.5	83.9	73.4	127.6	81.5	72.3	122.8	80.3	42.5	87.8	40.4	85.0	38.8	82.9	85.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 28.0	(b) 24.7	(c) 21.2	(e) -13.6	(f) 94.6	(g) 7.4	(h) 2.7	(i) -19.0	(j) 96.6	(k) -23.4	(l) 98.2	(m) -27.5	(n) 99.8	(o) -32.9	(p) 101.7		
(4) 28.0	(5) 24.7	(6) 21.2	(7) -15.0	(8) 81.4	(9) 6.9	(10) 2.1	(11) -19.9	(12) 82.9	(13) -23.9	(14) 84.1	(15) -27.8	(16) 85.2	(17) -32.8	(18) 86.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	49.3	20.9	25.0	37.1	49.1	60.9	70.3	73.5	71.0	64.3	51.8	39.1	27.0	
	DBStd	20.63	12.61	12.31	11.68	9.45	8.45	6.64	5.57	5.23	7.75	9.59	10.37	11.56	
	HDD50	3334	902	702	429	126	10	0	0	4	94	351	716		
	HDD65	6547	1367	1121	867	484	183	26	4	106	423	778	1178		
	CDD50	3074	0	1	28	99	349	608	729	651	433	149	24	3	
	CDD65	811	0	0	2	7	57	184	268	195	85	13	0	0	
	CDH74	7411	0	0	14	104	545	1737	2512	1573	799	126	1	0	
	CDH80	2399	0	0	1	20	140	613	907	460	232	26	0	0	
Wind	WSAvg	10.0	11.0	11.1	11.8	12.9	11.3	9.2	7.2	6.8	7.9	9.8	10.9	10.5	
Precipitation	PrecAvg	34.7	1.5	1.3	2.4	3.3	3.9	4.3	3.7	4.3	3.4	2.6	2.2	1.8	
	PrecMax	50.4	4.7	3.3	6.6	7.1	12.4	14.6	8.7	13.8	11.2	6.5	5.4	4.2	
	PrecMin	19.6	0.1	0.2	0.4	0.7	0.7	0.9	0.2	1.0	0.1	0.0	0.3	0.4	
	PrecStd	6.9	0.9	0.9	1.5	1.6	2.3	2.6	1.9	2.9	2.3	1.8	1.3	1.0	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.2	60.6	75.1	82.4	89.6	92.6	95.5	91.3	90.5	83.7	71.7	60.7	
		MCWB	48.0	51.7	62.2	63.1	71.4	74.5	79.4	76.5	73.6	71.6	59.3	56.8	
	2%	DB	45.4	52.0	66.3	76.7	83.6	90.1	90.8	88.2	85.9	77.3	64.0	53.4	
		MCWB	41.1	46.7	56.1	60.4	67.0	73.4	76.6	76.1	71.4	64.9	54.6	49.5	
	5%	DB	39.3	45.5	61.1	71.7	80.7	87.2	88.3	84.4	82.1	72.5	59.3	46.2	
		MCWB	36.4	41.1	53.0	57.5	65.4	72.0	75.5	74.1	68.8	61.6	52.1	42.8	
	10%	DB	36.4	39.4	54.5	65.9	75.6	83.2	84.4	81.9	79.1	67.7	54.8	41.5	
		MCWB	34.1	36.0	47.5	54.1	63.0	70.3	73.6	71.9	66.7	59.6	48.5	38.0	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.2	53.5	63.5	66.5	73.5	78.4	82.1	80.2	76.9	71.9	61.2	57.6
			MCDB	51.6	57.0	74.9	78.5	84.9	88.9	90.8	88.1	85.5	81.2	68.0	61.0
2%		WB	41.4	46.4	58.6	62.4	70.5	75.9	79.4	77.8	73.7	67.2	56.7	49.1	
		MCDB	45.4	50.5	65.0	73.3	80.0	85.6	88.6	85.6	81.9	74.2	62.4	52.1	
5%		WB	37.3	41.2	53.5	59.3	67.8	73.8	77.4	75.6	71.3	63.8	53.3	43.6	
		MCDB	39.8	45.4	60.4	69.2	76.8	83.2	86.3	82.6	79.4	69.8	58.7	46.7	
10%		WB	34.1	36.9	47.7	55.6	65.5	71.7	75.4	73.7	68.9	60.0	49.3	38.4	
		MCDB	36.3	39.7	54.3	64.9	73.8	80.9	82.7	80.3	76.4	66.6	54.9	41.5	
Mean Daily Temperature Range	5% DB	MDBR	15.7	16.5	18.5	21.6	21.2	20.4	20.1	19.8	22.8	20.9	18.7	15.1	
		MCDBR	18.4	21.0	25.8	30.1	24.9	23.3	22.2	21.7	25.7	25.7	24.2	19.8	
		MCWBR	15.1	16.2	17.1	17.8	13.6	11.2	12.0	11.9	13.3	15.4	16.9	16.1	
	5% WB	MCDBR	17.9	20.5	24.2	25.6	21.6	20.6	20.1	19.1	21.3	21.7	22.2	18.7	
		MCWBR	14.9	16.2	17.1	16.9	13.4	11.4	12.1	11.6	12.0	15.2	17.1	16.3	
Clear-Sky Solar Irradiance	taub	0.290	0.301	0.341	0.387	0.416	0.424	0.442	0.421	0.392	0.343	0.301	0.287		
	taud	2.404	2.384	2.371	2.255	2.239	2.263	2.200	2.259	2.312	2.451	2.502	2.432		
	Ebn at Noon	273	288	287	279	273	271	264	266	266	269	268	265		
	Edh at Noon	25	30	34	42	43	43	45	41	36	28	23	23		
All-Sky Solar Radiation	RadAvg	557	869	1184	1476	1731	1944	1931	1705	1435	941	622	455		
	RadStd	66	95	105	126	123	163	124	112	118	111	51	46		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (52 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page